RetailX-Store-Chain – Secure Azure Retail Architecture

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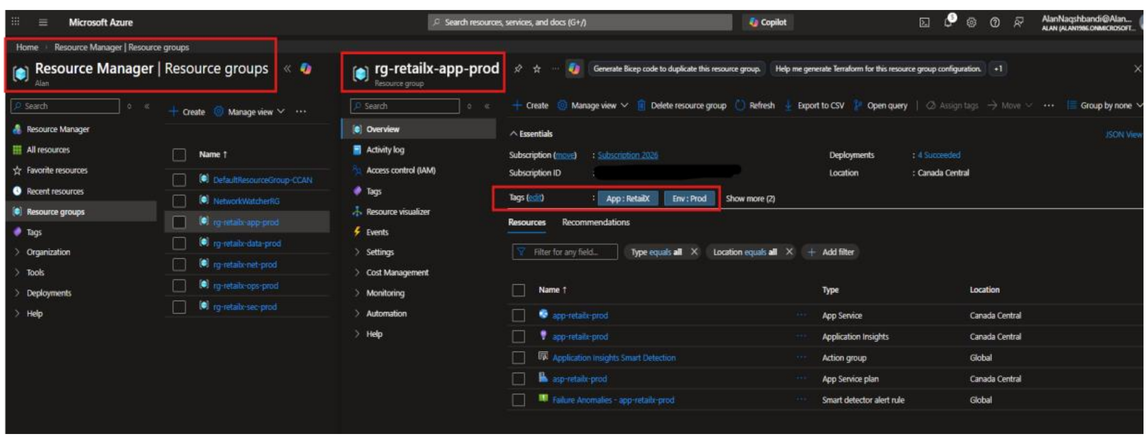
Executive Summary

This document provides structured evidence of the secure deployment and validation of the **RetailX Azure production architecture**. The solution implements a retail-grade design with network segmentation, Private Endpoints, identity-based access control, Defender for Cloud protections, and centralized monitoring. This architecture ensures secure backend services, streamlined operational visibility, and high availability, aligned with enterprise cloud standards.

PHASE 0 – Resource Organization

Purpose:
Establish governance, workload separation, and consistent tagging for cost tracking, ownership, and audit clarity.

- Created dedicated resource groups for network, app, data, security, and operations.
- Applied consistent tagging across all resource groups (App, Environment, Owner, and Data classification).

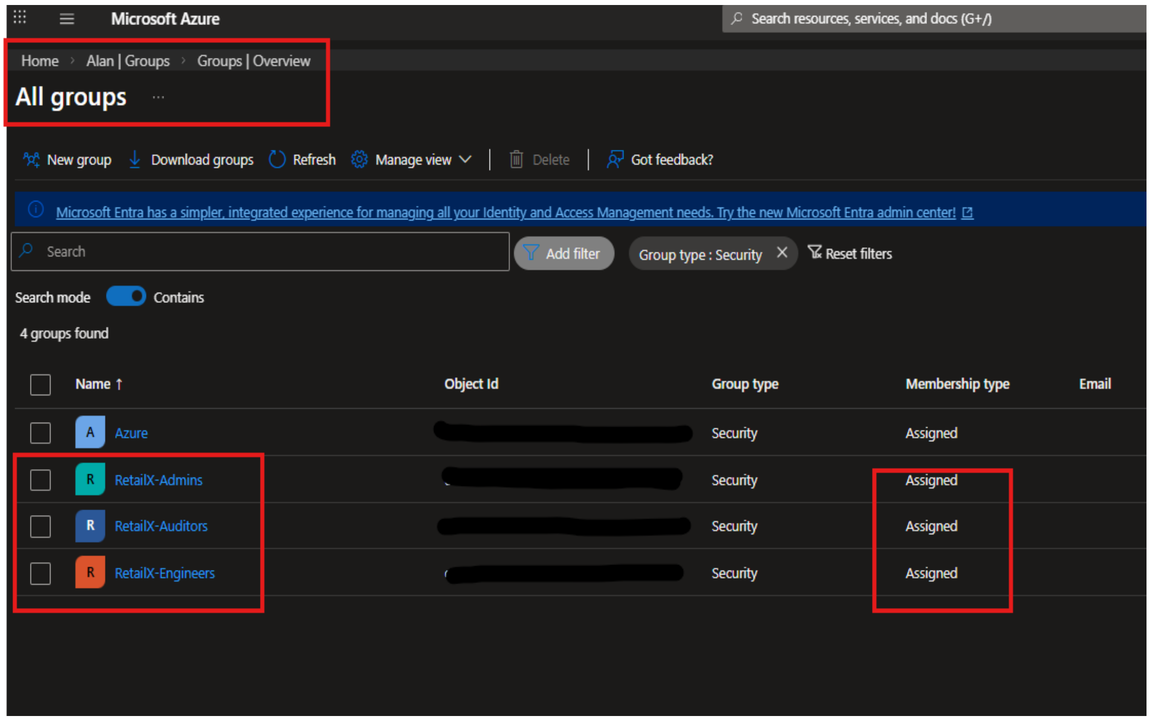


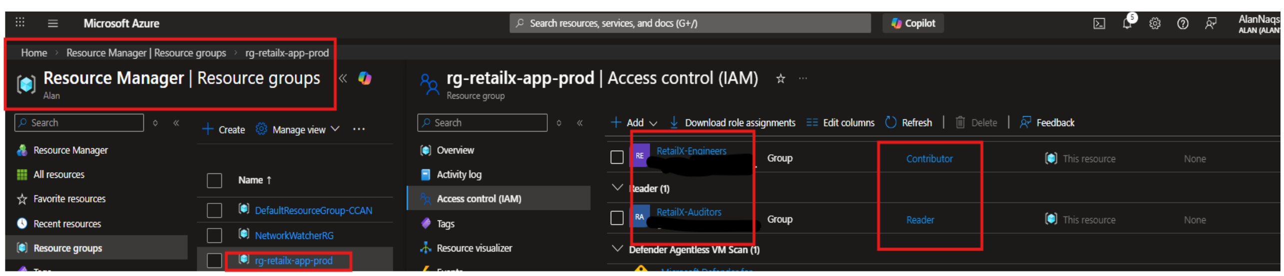
PHASE 1 – Identity & RBAC

Purpose:
Implement least-privilege access using Microsoft Entra ID security groups and RBAC at

the resource group scope, enforcing separation of duties and preventing direct user permissions.

- Create Microsoft Entra ID security groups: **RetailX-Admins**, **RetailX-Engineers**, **RetailX-Auditors**.
- Assign group-based roles (Owner for Admins, Contributor for Engineers, Reader for Auditors) at the appropriate resource group levels.



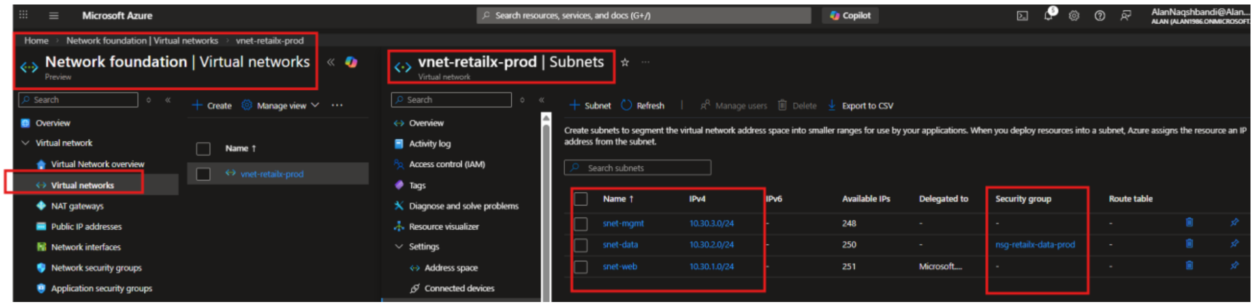


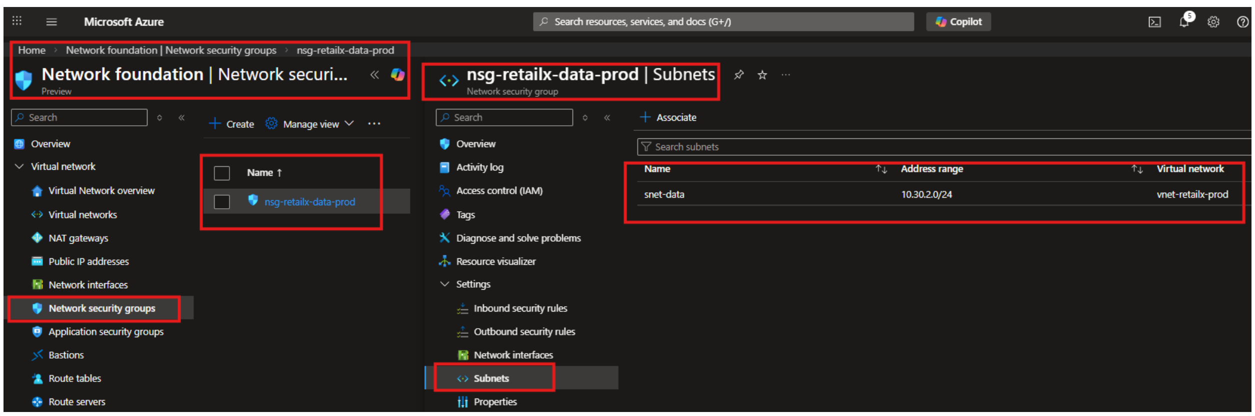
PHASE 2 – Network Segmentation & Private Connectivity

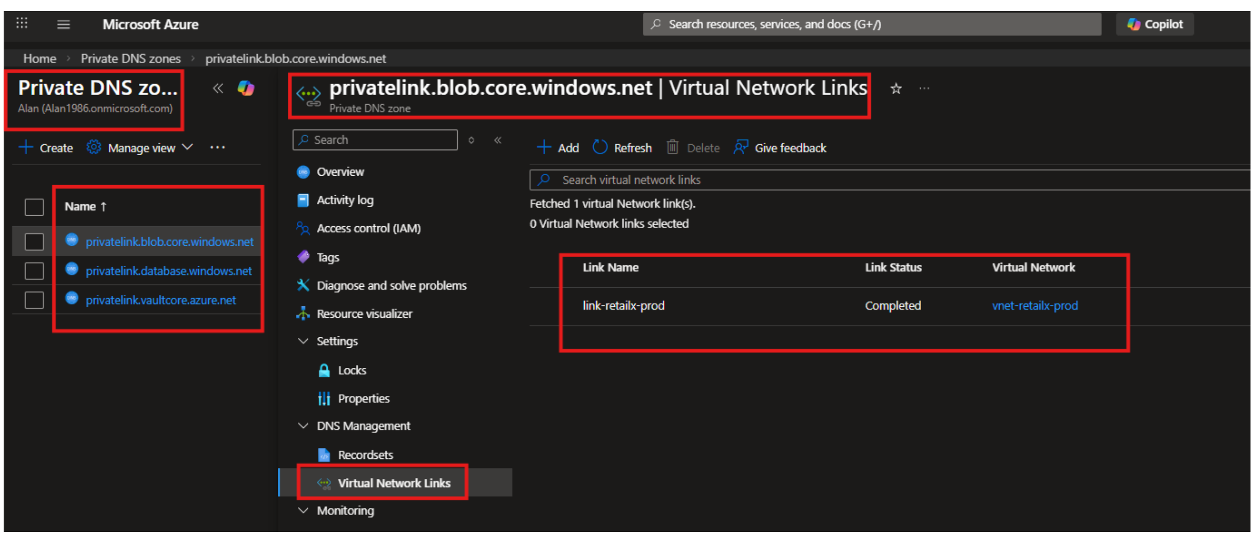
Purpose:
Build the network foundation for a “public frontend / private backend” model using a segmented VNet, subnets, and Private DNS integration to support Private Endpoints securely.

- Deploy Virtual Network (vnet-retailx-prod) with segmented subnets (Web, Data, Management).

- Apply Network Security Groups (NSG) to enforce private subnet isolation.
- Create and configure Private DNS zones (privatelink.database.windows.net, privatelink.blob.core.windows.net, privatelink.vaultcore.azure.net).
- Link DNS zones to the VNet.







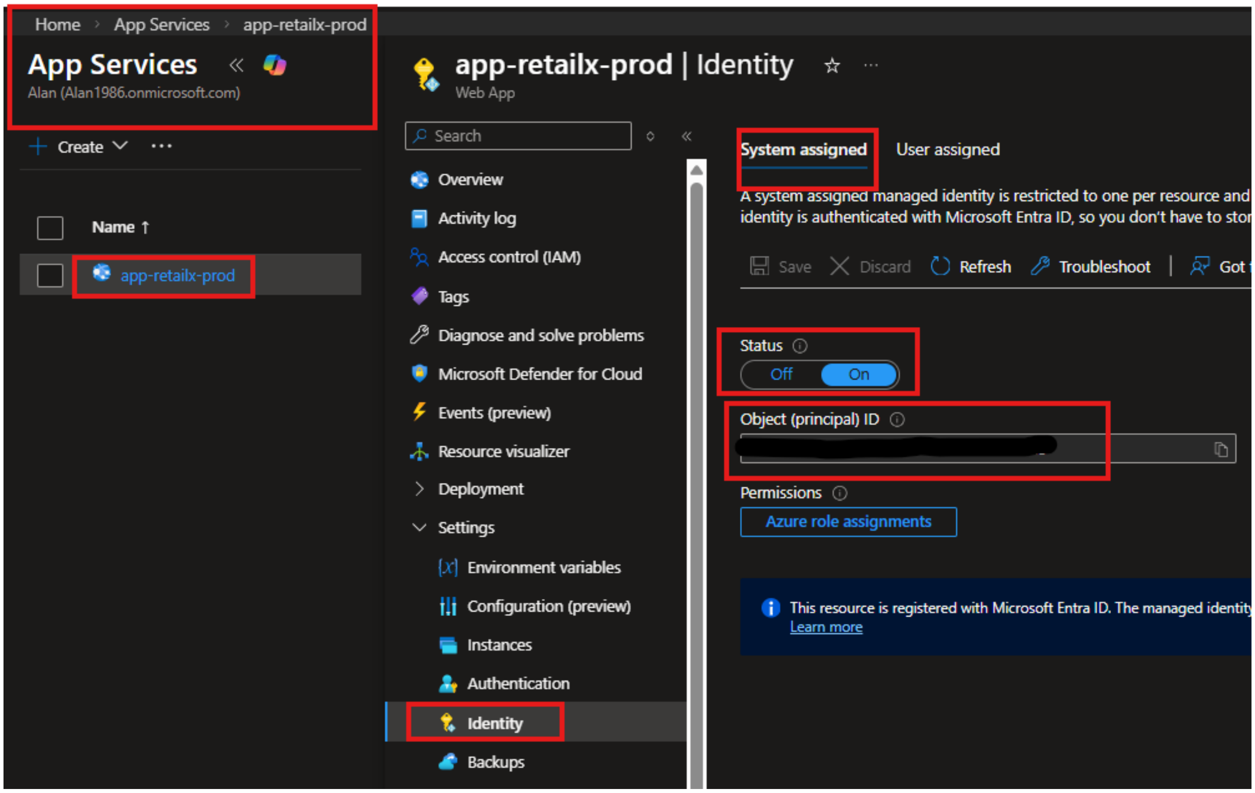
PHASE 3 – Application Layer (App Service)

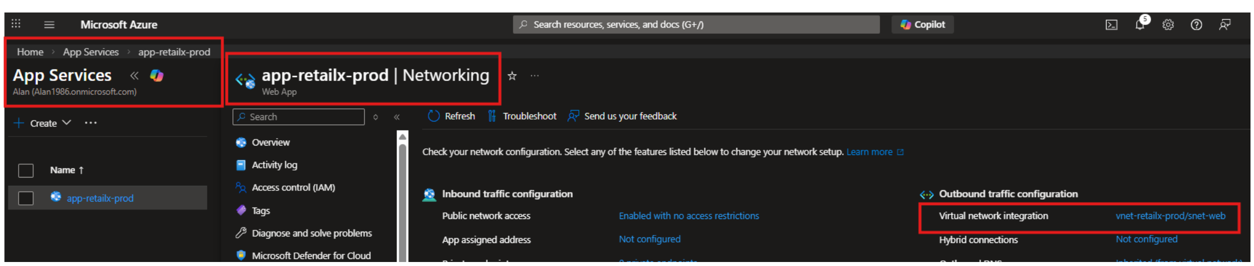
Purpose:

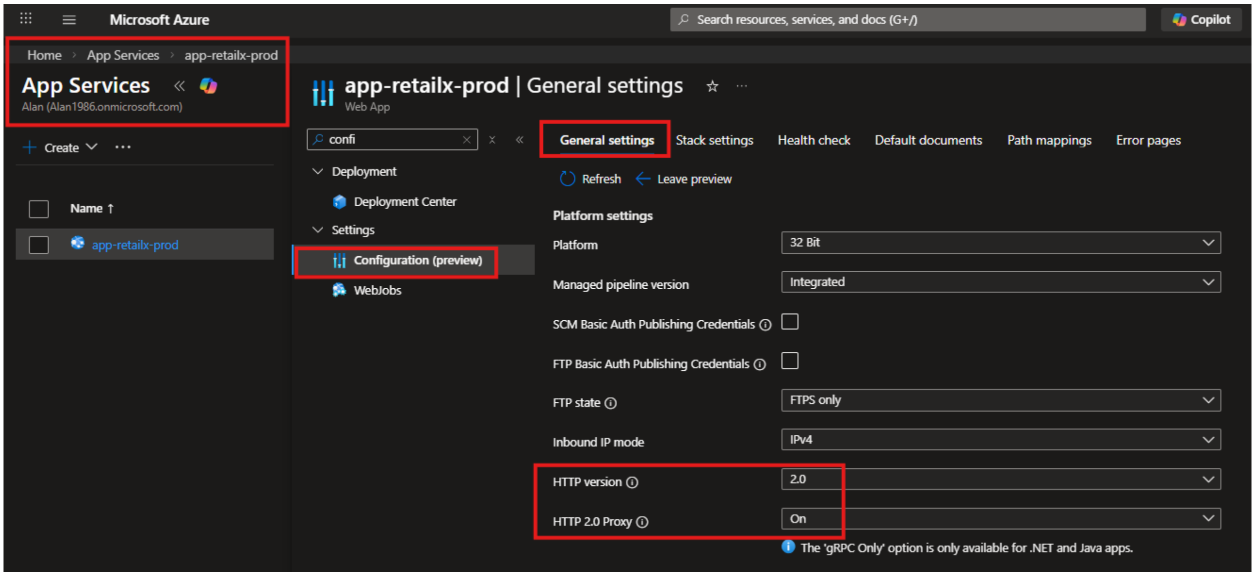
Deploy the public-facing application layer with secure defaults (HTTPS-only), enable

Managed Identity, and integrate the web app with the VNet to reach private backend services.

- Deploy Azure App Service on Premium tier (P1v3).
- Enable Managed Identity for secure authentication to backend resources.
- Integrate the App Service with the VNet for private backend access.
- Enforce HTTPS-only communication for the web app.



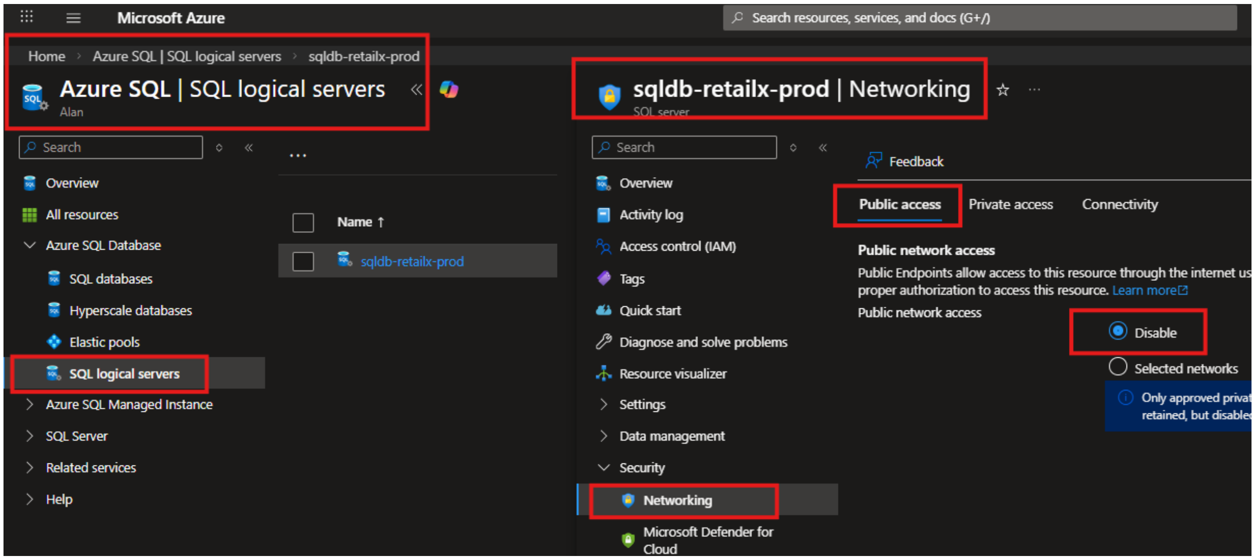




PHASE 4 – Secure SQL Deployment (Private + Protected)

Purpose:
Deploy the financial database with zero public exposure by disabling public access and enforcing private connectivity through Private Endpoint + Private DNS; enable Defender protections for database threat detection.

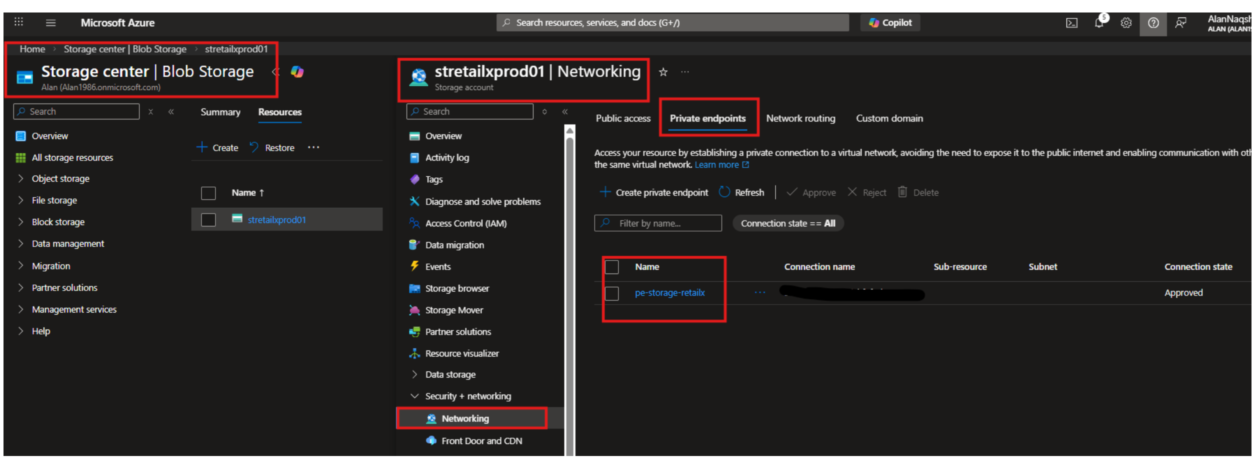
- Deploy Azure SQL Database (General Purpose tier).
- Disable public network access to the SQL Server.
- Configure Private Endpoint to ensure SQL Database is accessible only within the VNet.
- Attach Private DNS for SQL Server resolution.
- Enable Defender for SQL protection.

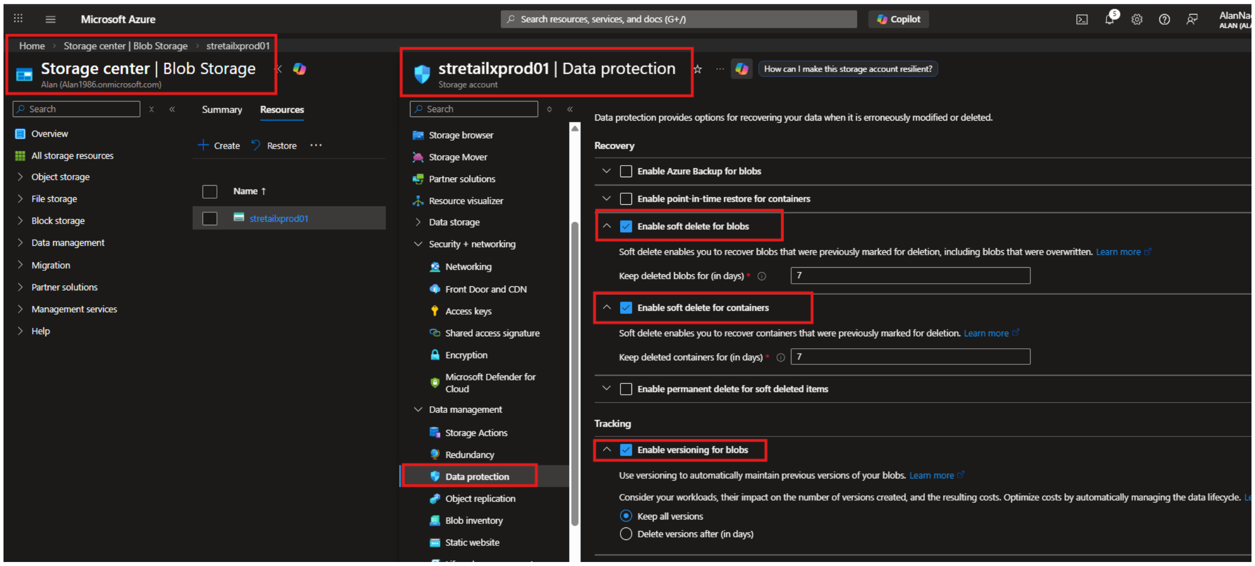


PHASE 5 – Secure Storage Deployment (Private + Data Protection)

Purpose:
Deploy secure storage for retail documents with public access disabled, private connectivity via Private Endpoint, and resilience controls (soft delete + versioning); enable Defender protections for storage threat detection.

- Create Azure Storage Account with LRS redundancy.
- Disable public access to the storage account.
- Configure Private Endpoint for secure data access.
- Enable Blob Soft Delete and Versioning for data protection.
- Enable Defender for Storage protection.

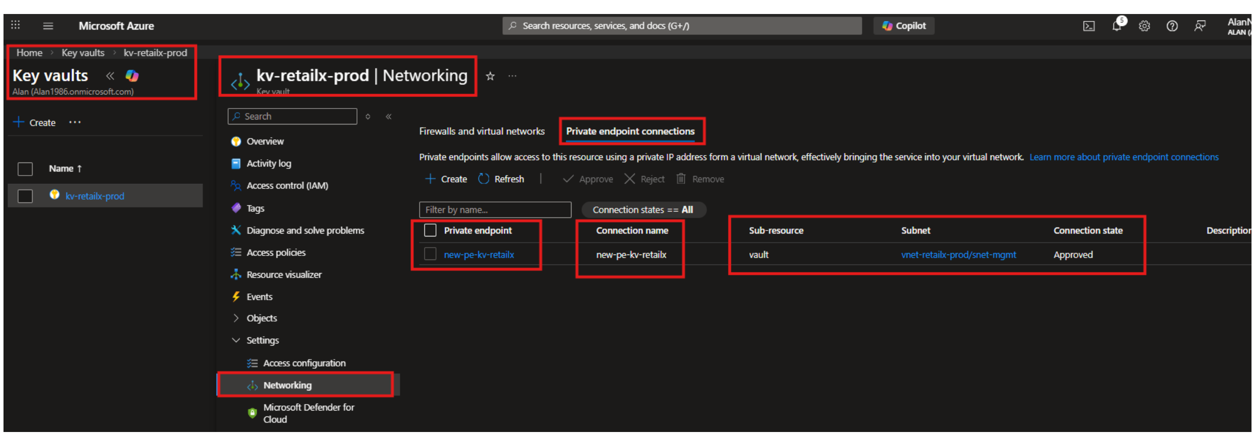


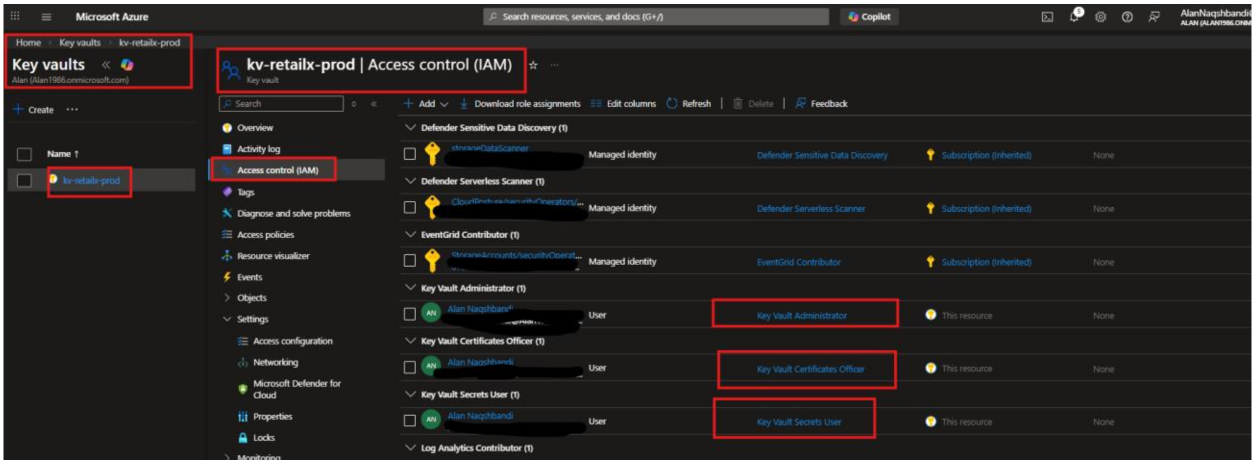


PHASE 6 – Secrets Management (Key Vault + Managed Identity)

Purpose:
Protect secrets using a private Key Vault (no public access), RBAC-based authorization, and Managed Identity for application access; store sensitive configuration such as database connection strings securely.

- Deploy Azure Key Vault with RBAC model.
- Disable public access to the Key Vault.
- Create Private Endpoint for secure access.
- Assign Managed Identity access to Key Vault for secure secrets management.
- Store sensitive secrets such as database connection strings.



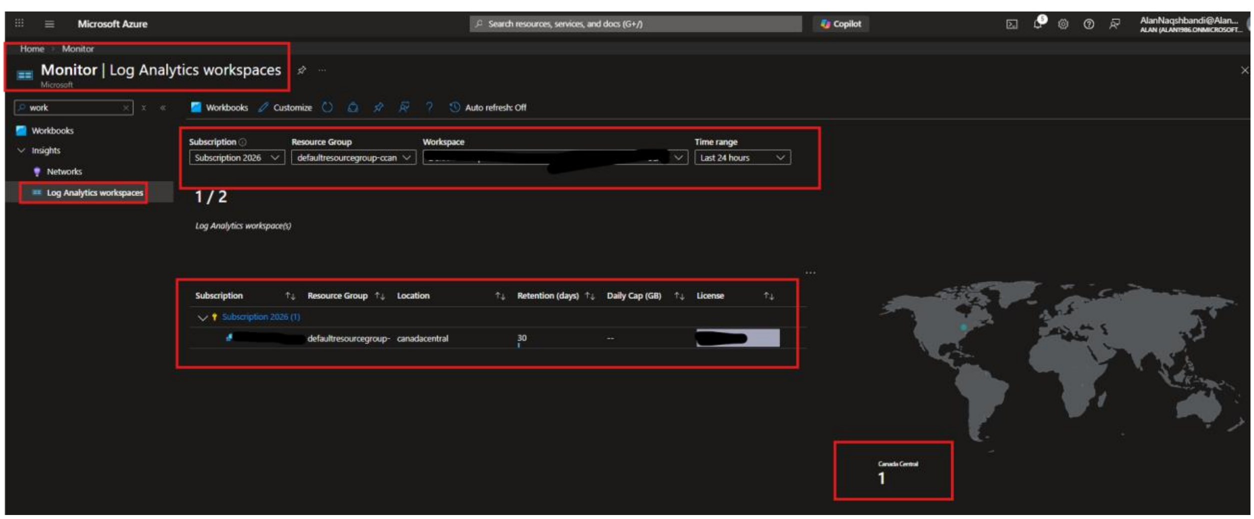


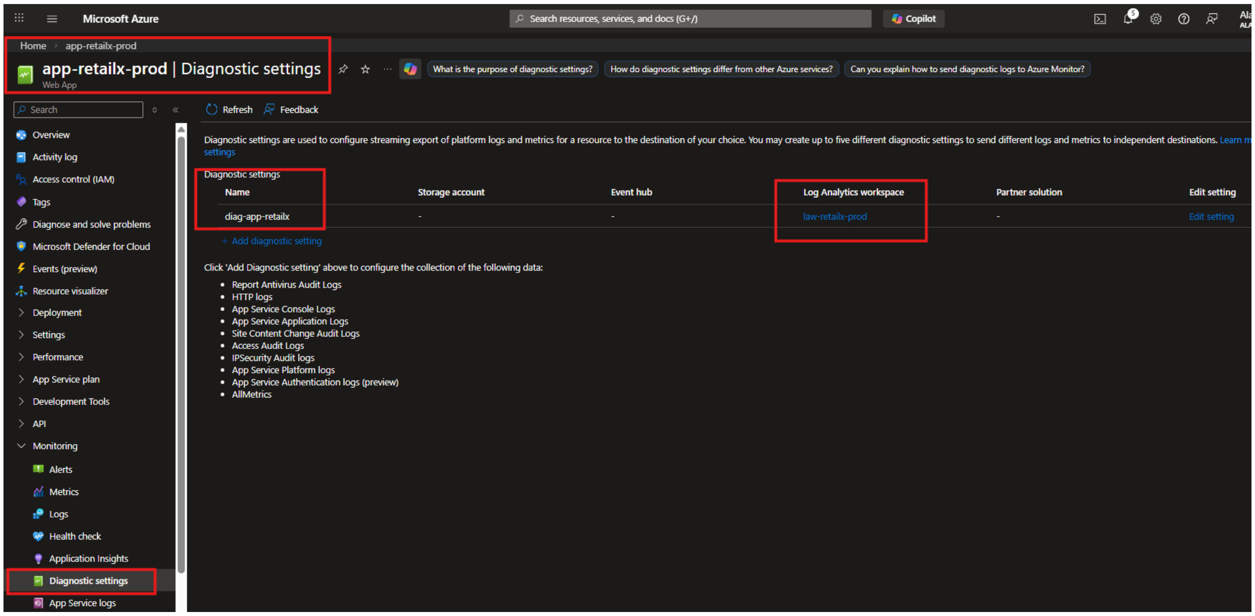
PHASE 7 – Monitoring & Threat Detection

Purpose:

Enable centralized logging and operational alerting for production visibility.

- Create Log Analytics Workspace for centralized log aggregation.
- Enable diagnostic settings for all resources (App Service, SQL, Storage, Key Vault).
- Configure alert rules for service availability, SQL CPU utilization, and storage anomalies.
- Enable Microsoft Defender for Cloud to continuously monitor and improve the security posture.



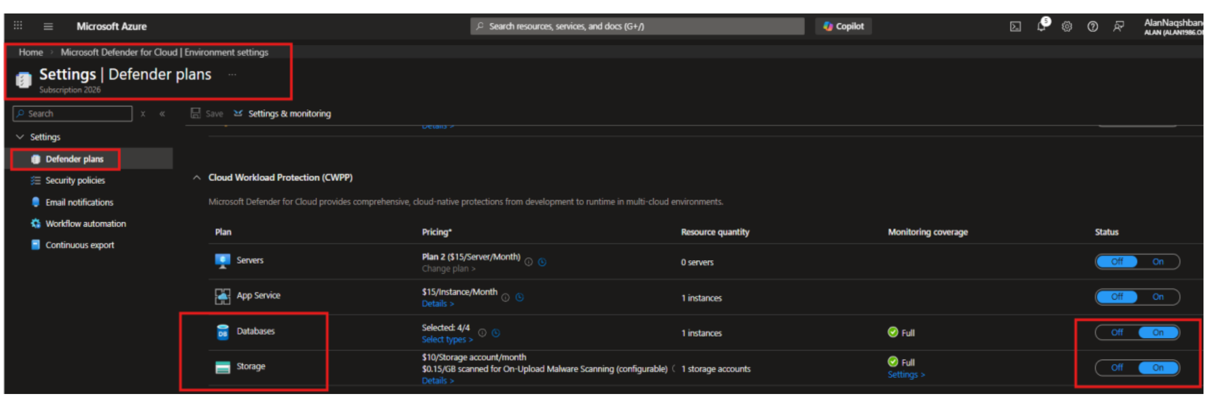


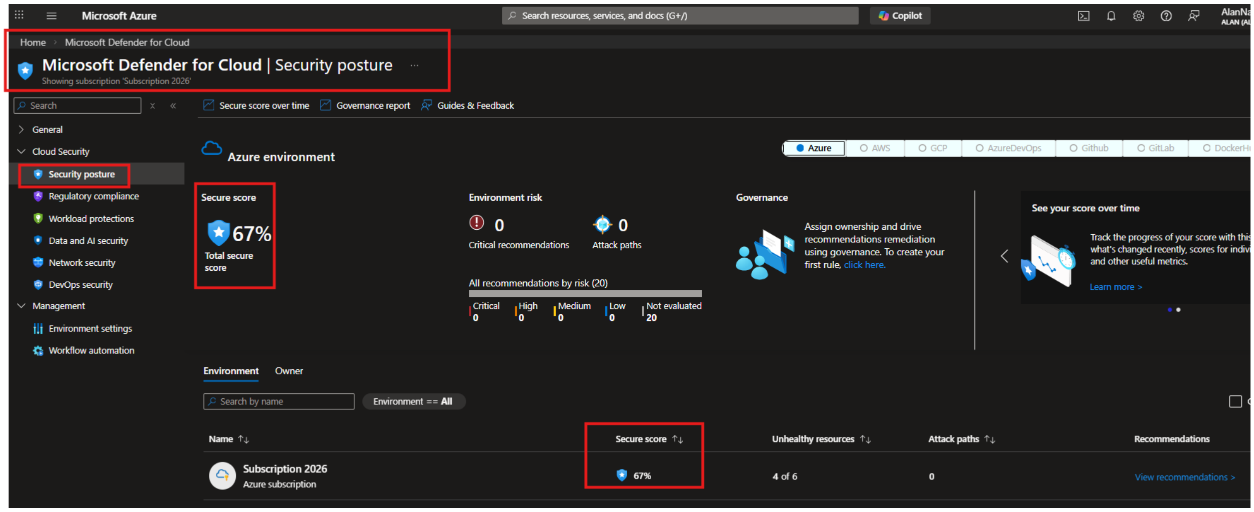
PHASE 8 – Production Validation

Purpose:

Validate the end-to-end security posture by confirming no public exposure of backend services, verifying Private Endpoint approvals and DNS resolution, validating identity-based access paths, and confirming monitoring and alerts are operational.

- Verify that all backend services (SQL, Storage, Key Vault) are fully private and not accessible from the public internet.
- Confirm private endpoint connectivity and DNS resolution for all private resources.
- Validate that Managed Identity is functioning correctly with all integrated services.
- Confirm that diagnostic settings, alert rules, and Defender for Cloud are operational.





Project Journey

This project demonstrates the end-to-end deployment of a secure, production-grade Azure environment designed for a multi-location retail organization. It implements structured governance (resource groups, tagging, and RBAC), a segmented network foundation with private-only backend services, private connectivity via Private Endpoints, identity-based access via Managed Identity, and centralized monitoring with Log Analytics and Azure Monitor. Security posture was strengthened using Microsoft Defender for Cloud protections for key services and validated through a final production review to confirm private exposure controls, access paths, and operational readiness.